

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, Java, Bash scripting

Tools/Frameworks: PyTorch, TensorFlow, JAX, FastAPI, Docker, Kubernetes, AWS SageMaker, TensorRT

Competence: Machine Translation, Large Language Models (LLMs), Neural Machine Translation, Natural Language Processing, Sequence-to-Sequence Models, MLOps, Deep Learning, AI/ML Systems at Scale

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and optimized deep learning models for metrological applications, enhancing model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to improve model accuracy and stability.
- Collaborated with PhD researchers to co-author a publication presented at the FLOMEKO conference.
- Developed dashboards for real-time data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Achieved 94% accuracy in vitiligo segmentation and improved processing speed by 5x using advanced AI algorithms, deployed via Docker and FastAPI.
- Developed a real-time tennis foul detection system leveraging TensorRT, reducing compute cost and time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure for scalable deployments.
- Designed and delivered educational sessions on machine learning and big data at universities, enhancing knowledge sharing.
- Developed proof-of-concept solutions using YOLO, few-shot learning, and CNNs for client demonstrations.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack web application providing AI-based services using Flask, HTML, and CSS.

- Built an AI chatbot application leveraging OpenAI API and large language model (LLM) frameworks.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented web scrapers and ETL pipelines for structured and unstructured data.
- Automated data ingestion and cleaning workflows for client analytics use cases, improving data processing efficiency.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Developed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation.
- Enhanced model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Designed and implemented a real-time Bitcoin price forecasting system using machine learning models and time-series analysis.
- Integrated data pipelines for real-time data ingestion and processing to enable accurate predictions.

Saaz: AI Self-Playing Chitrali Sitar

2023

Signal Processing, Music Generation

- Built the world's first AI-driven self-playing Chitrali Sitar, combining signal processing techniques and music generation LLMs.
- Project featured on Coke Studio in a music video by renowned artist Hassan Raheem.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Focused on advanced topics in Machine Learning, Natural Language Processing, and Large Language Models.
- Engaged in research on multilingual tasks and neural machine translation.

National University of Computer and Emerging Sciences **Aug 2018 – June 2022**
Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor for academic excellence.
- Specialized in Deep Learning, Computer Vision, and Data Engineering.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python